Aurora E-5 High-Temperature Failure: Safety Review

Prepared by Lena Brandt (Safety Officer) on 2026-03-24. Reviewed by Dr. Priya Raman.

1. Incident Summary

Cell AUR-C103 (electrolyte E-5) failed at cycle 412 during 45 °C cycling. The pouch swelled by 14% and vented without flame. The cell was moved to the safety cabinet within 10 minutes and no one was exposed.

2. Gas Analysis

GC-MS of the pouch gas found CO2 and ethylene (C2H4), consistent with decomposition of ethylene carbonate (EC) in the E-5 electrolyte at elevated temperature. No manufacturing defect was found: the seal integrity tests on the same batch passed.

3. Swelling Measurements

Cell	Electrolyte	Temp (°C)	Cycles	Swelling (%)
AUR-C103	E-5	45	412	14.0
AUR-C104	E-5 + FEC	45	800	3.2
AUR-C105	E-7	45	600	2.1
AUR-C106	E-7	45	600	2.4

All cells except AUR-C103 meet the 5% swelling limit defined in the test thresholds.

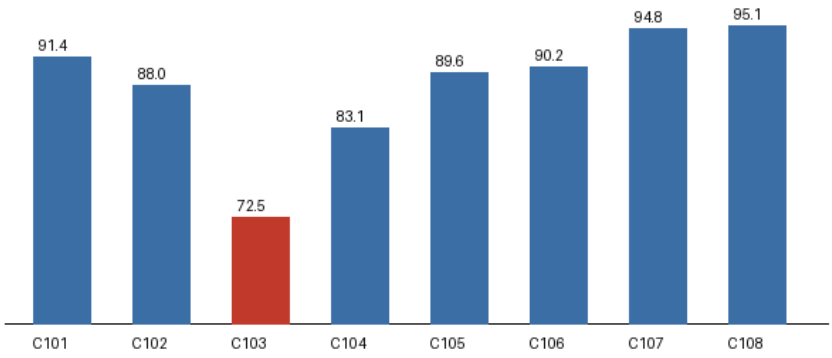


Figure 1: Capacity retention at the latest checkpoint, AUR-C101 to AUR-C108.

4. Recommendations

- Stop 45 °C testing with E-5 (already decided on 2026-03-12).
- Continue the E-7 qualification with swelling checks every 100 cycles.
- Add a gas-sampling port to pilot-line pouches before milestone M3.

5. Approval

Continuation of E-7 testing approved by Dr. Priya Raman on 2026-03-25.